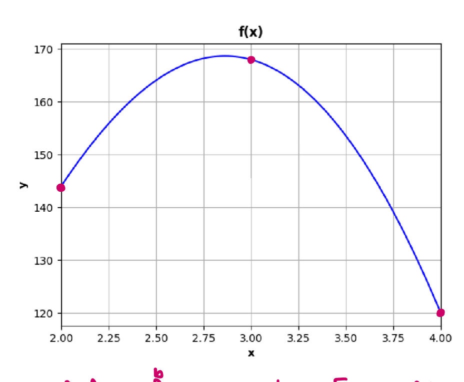
Quiz 3

11/24/25 13:30

1. **(Numerical Integration)** Approximate the following integral using Simpson's rule and the Trapezoid rule. Use the same three points for the Simpson's rule and Trapezoid rule (using 2 trapezoids). **Show your work.**

$$\int_2^4 f(x)dx$$



Trapezoid: $\int_a^b f(x)dx \approx \frac{1}{6}\Delta x\left[f(a)+f\left(\frac{a+b}{2}\right)+f(b)\right]$

Simpson: $\int_a^b f(x)dx \approx \frac{1}{2}\Delta x \sum_{i=1}^{n-1}\left[f(x_i)+f(x_{i+1})\right]$

$h = \frac{4-2}{2} = \frac{2}{2} = 1$

Trapezoid: $\int_2^4 f(x)dx \approx \frac{1}{2}\left[f(2)+2f(3)+f(4)\right]$

$= 0.5\left[144+2(168)+120\right] = 0.5\left[600\right] = \boxed{300}$

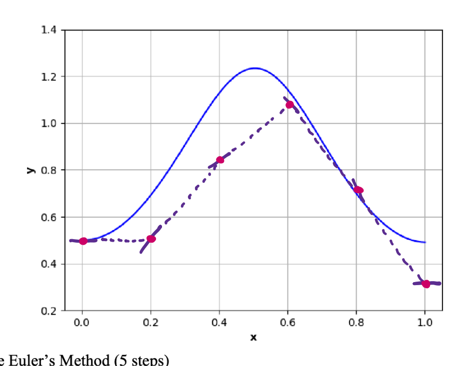
Simpson: $\int_2^4 f(x)dx \approx \frac{1}{3}\left[f(2)+4f(3)+f(4)\right]$

$= \frac{1}{3}\left[144+4(168)+120\right] = \frac{1}{3}\left[936\right] = \boxed{312}$

2. **(Explicit Integration)** Solve the following differential equation using explicit integration:

$$f(x,y) = \frac{dy}{dx} = \sin(2\pi x)\cdot e^y$$

given the initial condition $(x,y) = (0, 0.5)$ and $\Delta x = 0.2$. **Plot** the resulting function alongside the real result below. You need to plot all the calculated (x_i, y_i) in the figure and draw the slope $\left(\frac{dy}{dx}\right)$ at each (x_i, y_i).



Use Euler's Method (5 steps)

x	y	$f(x,y)$
0	0.5	0
0.2	0.5	1.568
0.4	0.814	1.326
0.6	1.079	-1.729
0.8	0.733	-1.980
1.0	0.337	0

① Solve for y w/ given DE

$\frac{1}{e^y}\cdot\frac{dy}{dx} = \sin(2\pi x)\cancel{e^y}\cdot dx \rightarrow \int e^{-y}dy = \int \sin(2\pi x)dx$

$-e^{-y} = -\frac{\cos(2\pi x)}{2\pi}+c \rightarrow c = \frac{\cos(2\pi x)}{2\pi} - e^{-y}$

$c = \frac{\cos(2\pi\cdot 0)}{2\pi} - e^{-0.5} = \frac{1}{2\pi} - e^{-0.5}$

$-1\left(-e^{-y} = -\frac{\cos(2\pi x)}{2\pi} + \frac{1}{2\pi} - e^{-0.5}\right)$

$\cancel{\ln}(e^{-y}) = \ln\left(\frac{\cos(2\pi x)}{2\pi} - \frac{1}{2\pi} + e^{-0.5}\right)$

$\boxed{y = -\ln\left(\frac{\cos(2\pi x)-1}{2\pi} + e^{-0.5}\right)}$ Exact

② Solve w/ Euler

Euler's Method:

$x_{n+1} = x_n + \Delta x,\ y_{n+1} = y_n + \Delta x\cdot f(x_n, y_n)$

$x_0 = 0,\ y_0 = 0.5$

$x_1 = 0+0.2 = 0.2$

$y_1 = 0.5+0.2(0) = 0.5$

$x_2 = 0.2+0.2 = 0.4$

$y_2 = 0.5+0.2(1.568) = 0.814$

$x_3 = 0.4+0.2 = 0.6$

$y_3 = 0.814+0.2(1.326) = 1.079$

$x_4 = 0.6+0.2 = 0.8$

$y_4 = 1.079+0.2(-1.729) = 0.733$

$x_5 = 0.8+0.2 = 1.0$

$y_5 = 0.733+0.2(-1.980) = 0.337$

3. **(Derivatives)** Calculate the gradient of the explicit function $\nabla\phi(-2,0)$ represented by the sample points in the grid using 2nd order finite differences. The gradient of a function is defined as its partial derivative along each axis:

$$\nabla\phi = \begin{bmatrix} d\phi/dx \\ d\phi/dy \end{bmatrix}$$

$\nabla\phi = \begin{bmatrix} d\phi/dx \\ d\phi/dy \end{bmatrix}$

y \ x	-4	-3	-2	-1	0	1	2	3	4
-3	3.2	2.1	1.8	0.8	1.0	0.8	1.8	2.1	3.2
-2	2.2	1.8	0.7	0.4	0.0	0.4	0.7	1.8	2.2
-1	2.1	1.4	0.4	-0.7	-1.0	-0.7	0.4	1.4	2.1
0	2.0	1.0	0.0	-1.0	-2.0	-1.0	0.0	1.0	2.0
1	2.1	1.4	0.4	-0.7	-1.0	-0.7	0.4	1.4	2.1
2	2.2	1.8	0.7	0.4	0.0	0.4	0.7	1.8	2.2

$f'(x) \approx \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}$

$\frac{\partial\Phi}{\partial x}(x,y) \approx \frac{\Phi(x+1,y) - \Phi(x-1,y)}{2(1)}$

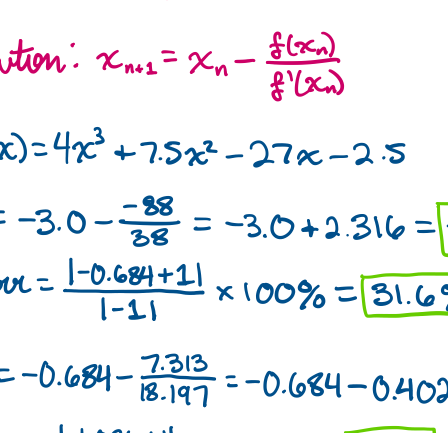
$\frac{\partial\Phi}{\partial y}(x,y) \approx \frac{\Phi(x,y+1) - \Phi(x,y-1)}{2(1)}$

$\frac{\partial\Phi}{\partial x}(-2,0) \approx \frac{\Phi(-1,0) - \Phi(-3,0)}{2} = \frac{-1.0-1.0}{2} = \boxed{-1.0}$

$\frac{\partial\Phi}{\partial y}(-2,0) \approx \frac{\Phi(-2,1) - \Phi(-2,-1)}{2} = \frac{0.4-0.4}{2} = \boxed{0}$

$\boxed{\nabla\Phi(-2,0) = \begin{bmatrix} -1.0 \\ 0 \end{bmatrix}}$

4. **(Roots of Equations)** Newton's Method. The following is a plot of the polynomial $p(x) = (x+5)(x-2.5)(x+1)(x-1)$:



- Calculate the root using **3 iterations** of Newton's method given $x_0 = -3.0$ given the expanded polynomial:

$$x^4 + 2.5x^3 - 13.5x^2 - 2.5x + 12.5$$

Fill the table below and include the **relative** error at each step (the real root is -1). At least Keep 3 significant digits. Illustrate how to determine the x_n on the figure also.

	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	E_r
0	-3.0	-88	38	-0.684	31.6%
1	-0.684	7.313	18.197	-1.086	8.6%
2	-1.086	-2.518	30.544	-1.004	0.4%
3	-1.004	-0.099	28.11	-0.9995	0.05%

Newton: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$f'(x) = 4x^3 + 7.5x^2 - 27x - 2.5$

$x_1 = -3.0 - \frac{-88}{38} = -3.0 + 2.316 = \boxed{-0.684}$

$\%err = \frac{|-0.684+1|}{|-1|} \times 100\% = \boxed{31.6\%}$

$x_2 = -0.684 - \frac{7.313}{18.197} = -0.684 - 0.402 = \boxed{-1.086}$

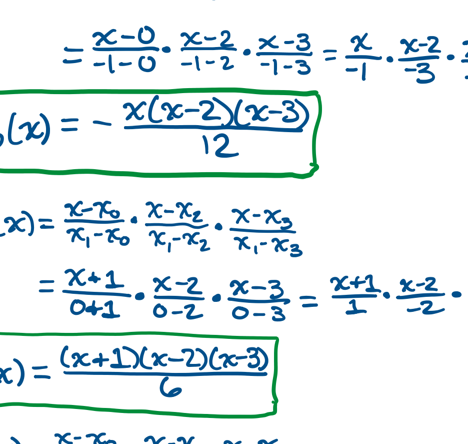
$\%err = \frac{|-1.086+1|}{|-1|} \times 100\% = \boxed{8.6\%}$

$x_3 = -1.086 - \frac{-2.518}{30.544} = -1.086 + 0.0824 = \boxed{-1.004}$

$\%err = \frac{|-1.004+1|}{|-1|} \times 100\% = \boxed{0.4\%}$

5. **(Interpolation)** A student is trying to use Lagrange Polynomials to fit a set of nodes (x_i, y_i), and has created the Lagrange Basis Polynomials in accordance with $\boldsymbol{x}$. If $\boldsymbol{y} = [-1, 3, 5, 2]$, please

- Show the expressions of all the Lagrange Basis Polynomials of this interpolation in production form.
- Show the Lagrange interpolation of the nodes. Please expand the polynomial.



$x = [-1, 0, 2, 3]$

$L_i(x) = \prod_{\substack{j=0 \\ j\neq i}}^{3} \frac{x-x_j}{x_i-x_j}$

$L_0(x) = \frac{x-x_1}{x_0-x_1}\cdot\frac{x-x_2}{x_0-x_2}\cdot\frac{x-x_3}{x_0-x_3}$

$= \frac{x-0}{-1-0}\cdot\frac{x-2}{-1-2}\cdot\frac{x-3}{-1-3} = \frac{x}{-1}\cdot\frac{x-2}{-3}\cdot\frac{x-3}{-4}$

$\boxed{L_0(x) = -\frac{x(x-2)(x-3)}{12}}$

$L_1(x) = \frac{x-x_0}{x_1-x_0}\cdot\frac{x-x_2}{x_1-x_2}\cdot\frac{x-x_3}{x_1-x_3}$

$= \frac{x+1}{0+1}\cdot\frac{x-2}{0-2}\cdot\frac{x-3}{0-3} = \frac{x+1}{1}\cdot\frac{x-2}{-2}\cdot\frac{x-3}{-3}$

$\boxed{L_1(x) = \frac{(x+1)(x-2)(x-3)}{6}}$

$L_2(x) = \frac{x-x_0}{x_2-x_0}\cdot\frac{x-x_1}{x_2-x_1}\cdot\frac{x-x_3}{x_2-x_3}$

$= \frac{x+1}{2+1}\cdot\frac{x-0}{2-0}\cdot\frac{x-3}{2-3} = \frac{x+1}{3}\cdot\frac{x}{2}\cdot\frac{x-3}{-1}$

$\boxed{L_2(x) = -\frac{x(x+1)(x-3)}{6}}$

$L_3(x) = \frac{x-x_0}{x_3-x_0}\cdot\frac{x-x_1}{x_3-x_1}\cdot\frac{x-x_2}{x_3-x_2}$

$= \frac{x+1}{3+1}\cdot\frac{x-0}{3-0}\cdot\frac{x-2}{3-2} = \frac{x+1}{4}\cdot\frac{x}{3}\cdot\frac{x-2}{1}$

$\boxed{L_3(x) = \frac{x(x+1)(x-2)}{12}}$

$P(x) = y_0L_0(x) + y_1L_1(x) + y_2L_2(x) + y_3L_3(x)$

$= -1L_0(x) + 3L_1(x) + 5L_2(x) + 2L_3(x)$

$= \frac{x(x-2)(x-3)}{12} + \frac{6(x+1)(x-2)(x-3)}{6\cdot 2} - \frac{2\cdot 5x(x+1)(x-3)}{2\cdot 6} + \frac{2\cdot x(x+1)(x-2)}{2\cdot 6}$

$= \frac{(x^2-2x)(x-3)}{12} + \frac{(6x+6)(x-2)(x-3)}{12} - \frac{(10x^2+10x)(x-3)}{12} + \frac{(2x^2+2x)(x-2)}{12}$

$= \frac{x^3-5x^2+6x}{12} + \frac{6x^3-24x^2+6x+36}{12} - \frac{10x^3-20x^2-30x}{12} + \frac{2x^3-2x^2-4x}{12}$

$\boxed{P(x) = \frac{-x^3-11x^2+38x+36}{12}}$

6. **(Least square)** Fit the following points to a cubic polynomial $p(x) = a + bx + cx^2 + dx^3$

x	-1.0	-0.5	0	0.5	1	1.5	2
y	0.1	2.2	3.1	2.7	1.9	1.9	3.1

please

7 points

- Show the cost function using least square fitting.
- Find the set of linear equations that can minimize the cost function.
- Solve the linear equations using Gaussian Eliminations to get a, b, c and d. Show your work.
- Predict the y at point $x = -1.5$ and point $x = 0.2$ respectively, using your cubic polynomial.

$\boxed{J(a,b,c,d) = \sum_{i=0}^{7}\left[y_i - (a + bx_i + cx_i^2 + dx_i^3)\right]^2}$ ← cost fcn

$\sum y_i = a\sum 1 + b\sum x_i + c\sum x_i^2 + d\sum x_i^3$

$\sum x_iy_i = a\sum x_i + b\sum x_i^2 + c\sum x_i^3 + d\sum x_i^4$

$\sum x_i^2y_i = a\sum x_i^2 + b\sum x_i^3 + c\sum x_i^4 + d\sum x_i^5$

$\sum x_i^3y_i = a\sum x_i^3 + b\sum x_i^4 + c\sum x_i^5 + d\sum x_i^6$

$\sum 1 = 7 \quad \sum x_i = 3.5 \quad \sum x_i^2 = 8.75$

$\sum x_i^3 = 11.375 \quad \sum x_i^4 = 23.1875$

$\sum x_i^5 = 39.59375 \quad \sum x_i^6 = 77.421875$

$\sum y_i = 15 \quad \sum x_iy_i = 11.1$

$\sum x_i^2y_i = 19.9 \quad \sum x_i^3y_i = 33.075$

$15 = 7a + 3.5b + 8.75c + 11.375d$

$11.1 = 3.5a + 8.75b + 11.375c + 23.1875d$

$19.9 = 8.75a + 11.375b + 23.1875c + 39.594d$

$33.075 = 11.375a + 23.1875b + 39.594c + 77.422d$

← set of linear equations

$$\begin{bmatrix} 7 & 3.5 & 8.75 & 11.375 \\ 3.5 & 8.75 & 11.375 & 23.1875 \\ 8.75 & 11.375 & 23.1875 & 39.594 \\ 11.375 & 23.1875 & 39.594 & 77.422 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 15 \\ 11.1 \\ 19.9 \\ 33.075 \end{bmatrix}$$

Eliminate bottom left triangle

$R2 = 2R2 - R1$

$$\left[\begin{array}{cccc|c} 7 & 3.5 & 8.75 & 11.375 & 15 \\ 0 & 14 & 14 & 35 & 7.2 \\ 8.75 & 11.375 & 23.1875 & 39.594 & 19.9 \\ 11.375 & 23.1875 & 39.594 & 77.422 & 33.075 \end{array}\right]$$

$R3 = R3 - 1.25R1$

$$\left[\begin{array}{cccc|c} 7 & 3.5 & 8.75 & 11.375 & 15 \\ 0 & 14 & 14 & 35 & 7.2 \\ 0 & 7 & 12.25 & 25.375 & 1.15 \\ 11.375 & 23.1875 & 39.594 & 77.422 & 33.075 \end{array}\right]$$

$R4 = R4 - 1.625R1$

$$\left[\begin{array}{cccc|c} 7 & 3.5 & 8.75 & 11.375 & 15 \\ 0 & 14 & 14 & 35 & 7.2 \\ 0 & 7 & 12.25 & 25.375 & 1.15 \\ 0 & 17.5 & 25.375 & 58.938 & 8.7 \end{array}\right]$$

$R3 = 2R3 - R2$

$$\left[\begin{array}{cccc|c} 7 & 3.5 & 8.75 & 11.375 & 15 \\ 0 & 14 & 14 & 35 & 7.2 \\ 0 & 0 & 10.5 & 15.75 & -4.9 \\ 0 & 17.5 & 25.375 & 58.9375 & 8.7 \end{array}\right]$$

$R4 = R4 - 1.25R2$

$$\left[\begin{array}{cccc|c} 7 & 3.5 & 8.75 & 11.375 & 15 \\ 0 & 14 & 14 & 35 & 7.2 \\ 0 & 0 & 10.5 & 15.75 & -4.9 \\ 0 & 0 & 7.875 & 15.1875 & -0.3 \end{array}\right]$$

$R4 = R4 - 0.75R3$

$$\left[\begin{array}{cccc|c} 7 & 3.5 & 8.75 & 11.375 & 15 \\ 0 & 14 & 14 & 35 & 7.2 \\ 0 & 0 & 10.5 & 15.75 & -4.9 \\ 0 & 0 & 0 & 3.375 & 3.375 \end{array}\right]$$

$3.375d = 3.375 \rightarrow \boxed{d = 1}$

$10.5c + 15.75d = -4.9$

$c = \frac{-4.9-15.75}{10.5} \rightarrow \boxed{c = -1.97}$

$14b + 14c + 35d = 7.2$

$b = \frac{7.2-35-14(-1.97)}{14} \rightarrow \boxed{b = -0.02}$

$7a + 3.5b + 8.75c + 11.375d = 15$

$a = \frac{15-11.375-8.75(-1.97)-3.5(-0.02)}{7} \rightarrow \boxed{a = 2.99}$

$\boxed{p(x) = x^3 - 1.97x^2 - 0.02x + 2.99}$

$\boxed{p(-1.5) = -4.79}$ $\boxed{p(0.2) = 2.92}$